

MedBLIP: A multimodal method of medical question-answering based on fine-tuning large language model

Lejun Gong^{a,*}, Jiaming Yang^a, Shengyuan Han^a, Yimu Ji^b

^a Jiangsu Key Lab of Big Data Security & Intelligent Processing, School of Computer Science, Nanjing University of Posts and Telecommunications, Nanjing 210023, China

^b Institute of High Performance Computing and Bigdata, Nanjing University of Posts and Telecommunications, Nanjing 210023, China

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ABSTRACT

Medical visual question answering is crucial for effectively interpreting medical images containing clinically relevant information. This study proposes a method called MedBLIP (Medical Treatment Bootstrapping Language-Image Pretraining) to tackle visual language generation tasks related to chest X-rays in the medical field. The method combine an image encoder with a large-scale language model, and effectively generates medical question-answering text through a strategy of freezing the image encoder based on the BLIP-2 model. Firstly, chest X-ray images are preprocessed, and an image sample generation algorithm is used to enhance the text data of doctor-patient question-answering, thereby increasing data diversity. Then, a multi-layer convolutional image feature extractor is introduced to better capture the feature representation of medical images. During the fine-tuning process of the large language generation model, a new unfreezing strategy is proposed, which is to unfreeze different proportions of the weights of the fully connected layer to adapt to the data in the medical field. The image feature extractor is responsible for extracting key features from images, providing the model with rich visual information, while the text feature extractor accurately captures the essential requirements of the user's question. Through their synergistic interaction, the model can more effectively integrate medical images and user inquiries, thereby generating more accurate and relevant output content. The experimental results show that unfreezing 31.25 % of the weights of the fully connected layer can significantly improve the performance of the model, with ROUGE-L reaching 66.12 %, and providing a more accurate and efficient answer generation solution for the medical field. The method of this study has potential applications in the field of medical language generation tasks. Although the proposed model cannot yet fully replace human radiologists, it plays an indispensable role in improving diagnostic efficiency, assisting decision-making, and supporting medical research. With continuous technological advancements, the model's performance will be further enhanced, and its application value in the medical field will become even more significant. The algorithm implementation can be obtained from <https://github.com/JiminFohill/MedicalChat.git>.

1. Introduction

With the advancement of generative artificial intelligence models, question-answering chatbots are gaining popularity across various professional fields, particularly in the medical domain. Swain et al. (2022) pointed out that healthcare professionals have always strived to provide the best services to patients. However, due to their busy schedules, healthcare professionals may not always be able to promptly engage with patients and deliver services. Therefore, AI-assisted diagnostic chatbots can offer 24/7 service and immediate access to essential information. In exploring the frontier of automated analysis and detection

methods, this study draws on two recent pioneering works. First, a data-driven fusion framework based on an end-to-end deep convolutional neural network has achieved remarkable success in identifying human-induced pluripotent stem cell-derived endothelial cells in photomicrographs (Iqbal et al., 2025). This framework offers new perspectives and methodologies for cell recognition. Secondly, another study employed deep convolutional neural networks for the automatic multiclass classification of skin lesions using dermoscopic images (Iqbal et al., 2021). This advancement further enhances the development of automated skin lesion recognition technology and provides robust support for clinical diagnosis. Together, these two studies offer valuable

* Corresponding author.

E-mail address: glj98226@njupt.edu.cn (L. Gong).

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insights and theoretical foundations for our current research on automated analysis and detection methods. Chatbots powered by large language models (LLMs) have made significant advancements in various natural language processing (NLP) tasks (Ali et al., 2023), such as question answering (QA), text classification, and interactive dialogues. It is important to note that even in fields requiring extensive expertise, such as code review (Koshti et al., 2023), business and economics (Han et al., 2023), and medical diagnosis (Raiaan et al., 2024), large language models (LLMs) have achieved impressive success (Qiu et al., 2023). However, while LLMs excel in language understanding within the medical domain, they do not possess the capability to interpret visual patterns, such as images and videos. This limitation hinders effective communication between visual content and the models. In recent years, the Visual Question Answering (VQA) task has garnered significant attention from both academia and industry, demonstrating immense potential in fields such as biomedicine, news media, and healthcare (Rafi et al., 2022). This task encompasses two modalities: computer vision and natural language processing, facilitating intelligent reasoning for images and posed questions, thereby enabling automated answer generation. The resulting Visual Language Pre-training (VLP) models can effectively comprehend visual content and excel in various visual language pre-training tasks. For instance, CLIP (Radford et al., 2021), BLIP (Li et al., 2022), BLIP2 (Li et al., 2023) have garnered significant attention due to their exceptional performance in multimodal fusion and their ability to generate human-like responses. Despite their numerous advantages, these models are not specifically tailored for the medical field. When addressing specialized domains, they often lack a comprehensive understanding of medical knowledge and may exhibit unreliable language generation. Accurate comprehension of medical expertise is essential in this context.

In this paper, we focus on medical visual question answering, which aims to develop models capable of understanding medical textual data and utilizing medical visual content to generate accurate responses. Compared to everyday images, medical images often exhibit high similarity, making it challenging for existing visual question answering models to accurately distinguish between them. This limitation reduces their effectiveness in providing medical information and decision support. Therefore, a significant challenge in this study is to fine-tune large-scale visual language models using medical visual question answering data, enabling the model to effectively discern the differences between medical images and capture the relevance between these images and textual questions. To address this challenge and validate the feasibility of applying visual language models to the medical domain, This work proposes a general and computationally efficient method for medical visual language processing (MedBLIP). This study comprehensively constructs a multimodal question-answering model architecture. The main contributions include:

- (1) This paper proposes a mirroring sample generation algorithm that creates mirrored samples based on the textual data of the original dataset, using the original data as positive samples. This approach enhances the model's understanding of the relationship between images and text and improves the model's generalization ability by increasing the diversity of the dataset.
- (2) This paper introduces a novel image feature extractor and integrates it with the Align-Former model to combine medical image data with medical question-answer text data. By employing a more advanced feature extractor, the model's ability to understand and analyze medical images is significantly enhanced. Additionally, the incorporation of textual data further improves the model's performance and effectiveness.
- (3) This paper proposes a method named MedBLIP, it is a multimodal method of medical question-answering based on fine-tune of large language model. Through comparative experiments, the optimal fine-tuning strategy was identified. By meticulously fine-tuning the large language generation model, it was better aligned

with the linguistic characteristics and requirements of the medical domain. This process significantly optimized the model, enabling it to demonstrate superior performance in specific tasks. The fine-tuned model can more accurately capture the context, terminology, and question types unique to the medical field, thereby greatly enhancing its performance and effectiveness.

2. Related work

2.1. Visual language pretraining models

In recent years, research on Visual Language Pre-training (VLP) has made remarkable progress, leading to the development of increasingly large-scale pre-trained models that continuously drive advancements in various downstream tasks. The CLIP (Contrastive Language-Image Pre-training) model is a pioneering contribution in this field. CLIP establishes a direct correspondence between image content and natural language descriptions through large-scale image-text contrastive learning. Its core advantage lies in its generalization ability: the pre-trained CLIP model can perform a variety of zero-shot learning tasks on unseen datasets, thanks to its training on a diverse range of images and descriptions. Despite its strengths in cross-modal understanding and retrieval tasks, CLIP has notable limitations, including a lack of precision in specific fine-grained tasks and inconsistent performance when confronted with complex language or images. To address these challenges, researchers have introduced the BLIP (Bootstrapped Language Image Pre-training) model, which aims to enhance the co-learning between language and vision through more refined training mechanisms. The design philosophy of BLIP focuses on improving the model's performance in specific application scenarios by bolstering its fine-grained understanding and generation capabilities while preserving CLIP's exceptional generalization ability. BLIP significantly enhances the model's capacity to manage complex visual and linguistic content, particularly excelling in tasks such as image captioning and fine-grained visual question answering. This improvement is achieved through more effective language-image interaction mechanisms and refined training strategies. Following BLIP, its successor, BLIP-2, further broadens the model's capabilities and application scope. Building on the strengths of BLIP, BLIP-2 improves the quality of generation and adaptability to various tasks by incorporating more advanced model architectures and training techniques. These enhancements allow BLIP-2 to achieve new levels of proficiency in understanding and generation tasks while improving its ability to process diverse and intricate data. However, both models exhibit subpar performance in medical visual language question answering tasks, as they lack a sufficient understanding of specialized medical images and tend to produce answers with limited expertise.

2.2. Visual language pretraining models in the medical field

The Medical Visual Question Answering (Med-VQA) task has recently garnered significant attention (Gong et al., 2021). Liu et al. (2023a) conducted an innovative study in the field of medical visual language, developing a straightforward yet effective method to enhance the efficacy of contrastive learning. Specifically, they streamlined the similarity calculation between medical image reports, transforming it into a similarity assessment between the reports themselves. This approach categorizes the image-report pairs into positive, negative, and additional neutral groups. Such precise sample classification facilitates the construction of a more appropriate contrastive loss function. However, while this method has enhanced the effectiveness of contrastive learning to some extent, its limitation lies in its exclusive focus on the similarity calculation between pairs of medical image reports, without incorporating generation tasks. In practical applications, generation tasks are also crucial in the medical visual language domain, such as generating accurate medical reports and facilitating medical question

answering. Medical-VLBERT (Liu et al., 2021) is an advanced model designed to assist in medical diagnosis by identifying abnormalities in COVID-19 scan images. This model not only detects lesion areas within the images but also automatically generates detailed medical reports based on these identified lesions. This innovative application significantly enhances the efficiency and accuracy of medical diagnoses, particularly when addressing a large number of COVID-19 cases, by providing rapid and reliable support for healthcare professionals. Despite the significant progress that Medical-VLBERT has made in automatically generating medical reports, the model still has certain limitations. Firstly, its primary focus is on report generation rather than on medical question answering, which means it may not be able to directly address specific medical queries. VQA-Adapter (Liu et al., 2024) is an innovative solution aimed at reducing the training cost of the Med-VQA (Medical Visual Question Answering) model by introducing a lightweight adapter component. This adapter component is parameter-efficient, allowing only the adjustment of adapter parameters during training while keeping all parameters of the CLIP large-scale visual model frozen, thus significantly reducing computational resource consumption. However, it still has some limitations. Firstly, freezing the parameters of the CLIP model during training may hinder the model's performance on specific tasks, particularly those that necessitate fine-tuning for particular datasets. Research on rare diseases or uncommon images may be constrained by limited data, which could adversely affect the model's performance and generalization capabilities. To achieve consistently higher model performance, large-scale datasets are essential, containing examples of diverse organ types and a variety of questions. PubMedCLIP (Eslami et al., 2023) is an innovative solution for Medical Visual Question Answering (MedVQA), recognized for its exceptional universality and adaptability. The model can effectively process medical images from multiple organs, providing healthcare professionals with a robust tool for image analysis and diagnosis across various medical scenarios. While the model demonstrates strong generalization, its performance on specific tasks or particular types of medical images may require more in-depth analysis and validation. The

CNX-B2 (A et al., 2023) model combines CNN and Transformer architectures, leveraging the efficient visual feature extraction capabilities of CNNs alongside the powerful report generation abilities of Transformers. In contrast, MedBLIP focuses on vision-language question-answering tasks rather than solely generating radiology reports. Its question-answering capabilities can be extended to broader clinical applications, such as patient-doctor conversations.

Considering these studies collectively, it is evident that multimodal representation learning and applications in the medical field have received extensive research and exploration. However, more efforts are needed to address challenges in performance and stability to ensure the effectiveness of these models in practical medical applications. Due to the complexity of medical images and the obscurity of specialized terminology, these models still face challenges in understanding medical images and generating accurate answers.

3. Methods

BLIP-2 employs a strategy of freezing the image encoder to learn visual-language features, subsequently integrating the frozen large language model to generate responses. To tailor the model for the medical visual language generation task, this study introduces MedBLIP (Medical Treatment Bootstrapping Language-Image Pretraining), as illustrated in Fig. 1. The study fine-tunes specific components of the fully connected layers of the large language model while incorporating a new image feature extractor and utilizing the Align-Former fusion mechanism to merge medical image data with medical question-and-answer text data. Firstly, multi-layer convolutional neural networks are employed to extract features from chest X-ray images, which are then encoded by the image encoder. Meanwhile, original question-answer samples are mirrored using a mirroring algorithm to enhance data diversity. Subsequently, image encoding and text encoding are integrated through the Align-Former fusion mechanism, and the combined features are decoded into text output by the large language model. This section begins by introducing image processing and mirroring algorithms. This

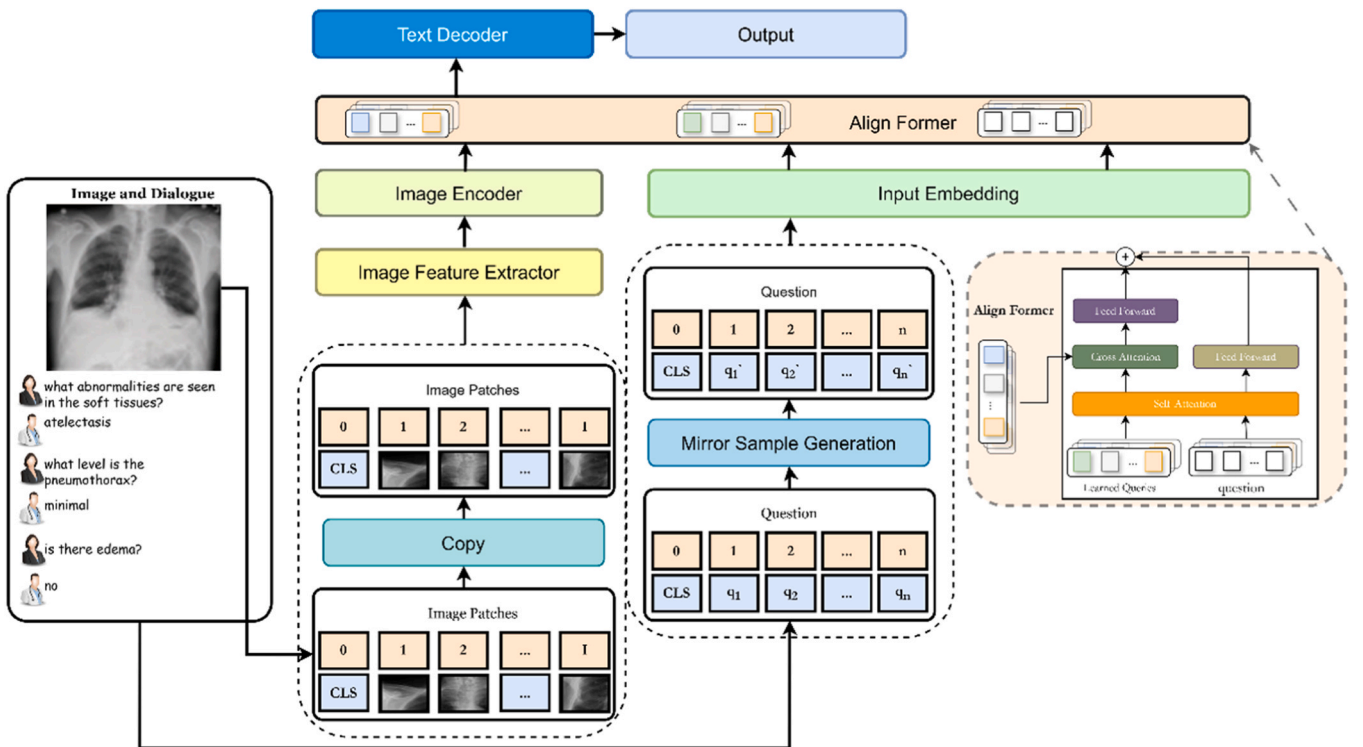


Fig. 1. MedBLIP schematic diagram. From left to right, the figure shows the image feature extractor, mirrored sample generation, and Align-Former for multi-modal fusion.

step ensures the effective processing of image data to meet the requirements of model training while enhancing data diversity and richness. The application of the mirroring algorithm increases the size of the dataset and improves the model's generalization capabilities. Next, the principles of the multi-layer image feature extractor are introduced. The design of this feature extractor aims to extract richer feature representations from medical images, thereby enhancing the model's understanding and analytical capabilities. By implementing a multi-layer feature extractor, various levels of features can be captured and effectively integrated. Finally, the fine-tuning process of the large language generation model is described. This process seeks to further optimize the model's performance on specific tasks. By fine-tuning the large language generation model, it can better adapt to the linguistic characteristics and task requirements of the medical domain, ultimately improving the model's performance and effectiveness.

3.1. Data processing and mirror sample generation

Firstly, this study focuses on processing chest X-ray images obtained from the MIMIC-CXR-JPG dataset in Table 1. Additionally, medical question-answer texts related to these images are selected from the Medical-Diff-VQA dataset. The primary objective of this study is to preprocess these images to ensure they can be effectively understood and utilized by the model. To achieve this goal, the image data is resized to a uniform dimension, typically (224, 224). This step ensures that the model can process input data at a consistent size, thereby enhancing the stability and effectiveness of the model. Next, the resized images are encoded using an image encoder to obtain image encoding matrices. This process converts the images into a digital format, allowing the computer to process and analyze them effectively. After obtaining the image encoding matrices, calculate the mean and standard deviation of these matrices. These statistics provide crucial information about the distribution of the image data, assisting the model in better understanding the characteristics and distribution of the data. Finally, use the following formula to determine the overall mean and standard deviation for all images:

$$I_{mean} = \frac{\sum_i^N mean(I_i)}{N} \quad (1)$$

$$I_{std} = \frac{\sum_i^N std(I_i)}{N} \quad (2)$$

N is the number of images, $mean()$ is the function to compute the mean of all elements, $std()$ is the function to compute the standard deviation of all elements. Then, normalize the images:

$$I_{result} = Image\ Processor(I_i, I_{mean}, I_{std}) \quad (3)$$

In this study, we apply a mirroring sample generation algorithm for data augmentation in doctor-patient question-and-answer texts. We begin by traversing the Medical-Diff-VQA dataset, which encompasses

various question types, question content, and doctors' answers. Initially, we classify the text using a question type classifier, followed by data generation through a mirroring sample generator, as illustrated in Fig. 2. For each data entry, we extract the question type and content, subsequently generating new questions and answers based on different question types. For questions that assess abnormality, we modify the questions and answers to create scenarios that are opposite to the original inquiries. Specifically, we transform affirmative questions into negative ones. In the case of existential questions, we adjust the questions and answers to produce contrasting scenarios, changing the answers from affirmative to negative. The generated data demonstrates a significant increase in volume compared to the original dataset.

Considering various enhancement algorithms, image-level data augmentation primarily includes techniques such as rotation, translation, scaling, flipping, cropping, and color transformation. These methods alter the visual content of images, allowing the model to learn more generalized features and enhancing its robustness to variations in image data. In contrast, mirrored sample generation algorithms—such as the reorganization of question-answer pairs and variations in sentence structure—focus on enriching the semantics and expression of textual data, thereby improving the model's understanding of medical terminology and question-answer logic. Compared to image augmentation, mirrored sample generation algorithms provide significant advantages in enhancing the model's language comprehension, increasing data diversity, reducing the risk of overfitting, streamlining annotation processes, conserving computational resources, and improving the model's adaptability in specific medical domains. These benefits enable question-answering systems to deliver clinical decision support more efficiently and accurately.

This study also attempted to use GAN (Goodfellow et al., 2014) models to augment the textual question-answer pairs in the dataset. However, the GAN-based QA training proved challenging, as it was difficult to balance the generator and discriminator or maintain proper control, resulting in poor-quality generated samples. In the GAN model experiments, most of the generated outputs were unusable. As shown in Table 2, issues such as syntactic and logical collapse, model misinterpretation, and confusion of technical terms were frequently encountered. For similar generative data augmentation tasks, this study suggests that employing LLMs (Large Language Models) would yield better results.

3.2. Image feature extractor

In the field of medical image analysis, there is often considerable similarity among images, evident in their shared structures and patterns. To address this characteristic, our study employs a transformer-based image embedding layer, which comprises 39 meticulously designed attention layers that effectively capture the deep-level features of images. This transformer-based embedding layer is not only capable of recognizing complex correlations within images but also adept at extracting highly abstract feature representations, thereby enhancing the model's comprehension of medical images.

As illustrated in Fig. 3, our study presents an advanced image feature extractor composed of three convolutional network layers, each utilizing a distinct patch processing strategy. The innovation of this design lies in its capacity to progressively expand the network's receptive field, thereby capturing a wider array of structures and subtle features within the images. Through this hierarchical feature extraction approach, our model can delve deeper into the intrinsic information of medical images, offering robust support for subsequent diagnosis and analysis. The extractor comprises three convolutional layers, each utilizing patches of different sizes. This design aims to progressively enlarge the receptive field, thereby enhancing the ability to capture diverse structures and features within the image. The extractor starts with a smaller patch size, typically 2×2 , and then gradually increases the patch size until reaching 16×16 , following the specific formula:

Table 1
GAN-based data enhancement effect analysis.

Raw data	GAN-based augmented data	Analysis of results
is there evidence of edema in this image?	is there any proof of edema visible here?	Success
is there evidence of hypoxemia in this image?	image this in evidence of hypoxemia is there?	The grammatical logic of the sentence collapses
is there pleural effusion in the right lung area?	is there liquid trouble in the right lung zone?	Insufficient comprehension of the model, 'effusion' translates directly to 'liquid trouble'
where in the image is the lung opacity located?	where in the image is the kidney cloudiness located?	Lack of modelling ability to consider 'kidney' and 'lung' as the same organ

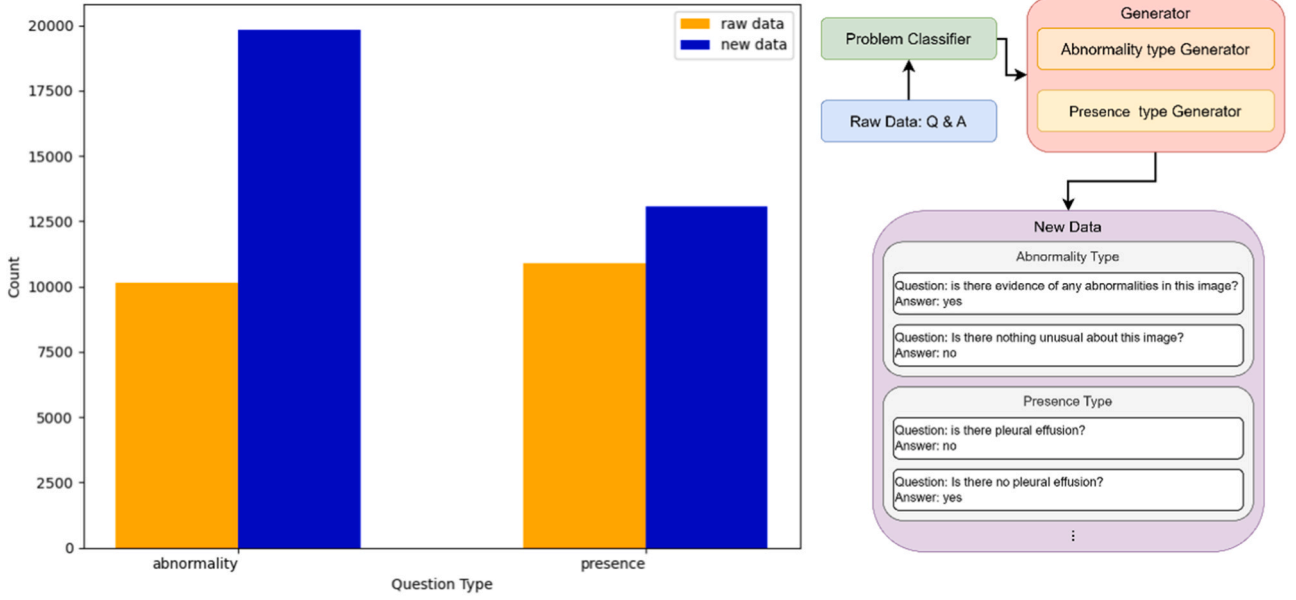


Fig. 2. Mirror sample generation process. The number of samples in the abnormality category doubled after data augmentation, while the presence category increased by 2500 samples. The right figure illustrates the specific process of data augmentation.

Table 2

Experimental dataset.

Data	Q&A	Images	Reports	URL
MIMIC-CXR-JPG	-	377,110	227,827	https://doi.org/10.13026/8360-t248
Medical Diff-VQA	700,703	164,324	-	https://doi.org/10.13026/5jes-bx23

$$I_1 = GELU(BN(W_{k=4,=4}^1 * I + b_1)) \quad (4)$$

$$I_2 = GELU(BN(W_{k=2,=2}^2 * I_1 + b_2)) \quad (5)$$

$$I_3 = GELU(BN(W_{k=2,=2}^3 * I_2 + b_3)) \quad (6)$$

$$W \sim U(-\sqrt{\frac{6}{I_{num} + O_{num}}}, \sqrt{\frac{6}{I_{num} + O_{num}}}) \quad (7)$$

where $I \in \mathbb{R}^{bt \times c \times h \times w}$, bt represents the batch size, c represents the number of channels, h and w are the height and width of the image, W represents the weights, b represents the bias, BN represents the batch normalization function, $GELU$ is the activation function, and U represents the Xavier initialization method.

This design enables the model to observe and analyze images at various scales, thereby facilitating a comprehensive understanding of the image content. In each convolutional layer, batch normalization is applied to the patches prior to the application of the non-linear activation function. This step accelerates the training process, mitigates issues such as gradient vanishing and exploding, and enhances the model's generalization capabilities. Batch normalization ensures that the inputs to each layer maintain similar distributions, making it easier for the model to learn effective feature representations. By incorporating such an image feature extractor, critical features can be more efficiently extracted from medical images, providing more accurate inputs for subsequent tasks. This feature extractor, specifically tailored to the characteristics of medical images, significantly improves the model's performance and effectiveness in the medical domain. This paper evaluates the performance of the transformer-based embedding layer in comparison to the multi-layer convolutional feature extractor in the experimental section, as presented in Table 3.

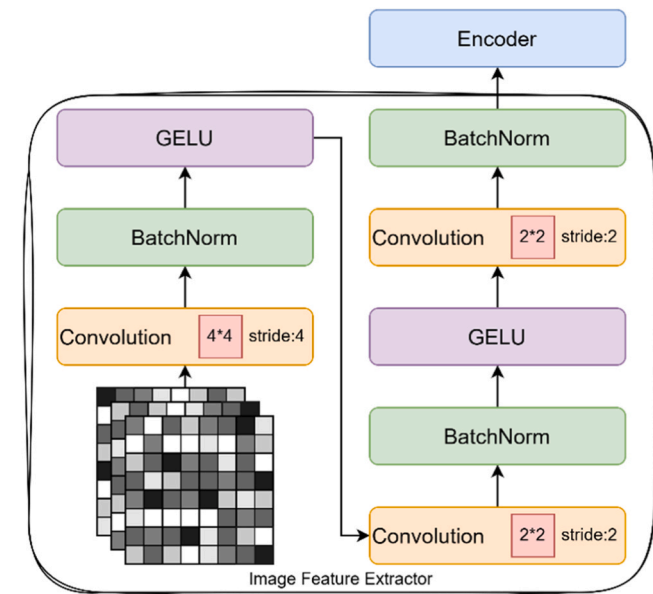


Fig. 3. Image feature extractor. Image features are extracted through multiple convolutional layers.

Table 3

Comparison of the effects of data augmentation algorithms.

Methods	B1(%)	R1(%)	RL(%)
MedBLIP-31.25	58.95	63.36	63.03
MedBLIP-31.25-AUG	62.42	66.21	65.90

3.3. Unfreezing the fully connected layers of the language model

The emergence of large language models has significantly impacted the field of natural language processing. These attention-based models have demonstrated remarkable capabilities in understanding and generating human dialogue contexts, surpassing most state-of-the-art models in the field. However, they are not suitable for domains that handle sensitive data, such as the medical field. Without further fine-tuning, the deployment of existing pretrained models is often not as effective for data in specific domains (Liu et al., 2023b). Performing full-scale fine-tuning of model weights and training models with billions of parameters necessitates costly GPU resources, which poses challenges for small laboratories and companies seeking to engage in this research field. LoRA (Hu, 2022) technology aims to adapt to specific tasks by introducing a limited number of trainable parameters while keeping the majority of the pretrained model parameters unchanged. This approach conserves computational resources, enables the model to swiftly adapt to new tasks, and mitigates overfitting issues that can arise from an excessive number of free parameters. However, existing research indicates that LoRA has not yet surpassed traditional full-parameter fine-tuning in many tasks (Dettmers et al., 2023). In this study, a novel approach was implemented to enhance the adaptation of multimodal models to medical data by unfreezing a portion of the language model's weights within the multimodal framework. The research focused on unfreezing varying proportions of the fully connected layer weights in the language model. In language models with 32 decoder layers, fully fine-tuning all parameters demands substantial computational resources. To mitigate resource consumption, this study proposes five training strategies that involve unfreezing 18.75 %, 25 %, 31.25 %, 37.5 %, and 43.75 % of the fully connected layer weights, as illustrated in Fig. 4. These strategies not only enhance model performance but also reduce excessive utilization of computational resources. A comparison of the five strategies reveals that unfreezing 31.25 % of the fully connected layer weights yields the best results. This approach offers a viable solution for delivering more accurate and efficient responses in the medical domain.

In Fig. 4, MedBLIP-18.75 refers to unfreezing the fully connected layers in the first three and last three decoder layers. MedBLIP-25 corresponds to unfreezing the fully connected layers in the first four and last four decoder layers, while MedBLIP-31.25 involves unfreezing the fully connected layers in the first five and last five decoder layers. Similarly, MedBLIP-37.5 and MedBLIP-43.75 represent unfreezing the fully connected layers in the first six and last six layers, and the first seven and last seven layers, respectively. A comparison of the five strategies was

conducted by training and testing them on the same dataset. The experimental results revealed that MedBLIP-31.25 achieved the best performance, demonstrating that unfreezing the fully connected layers in the first five and last five decoder layers yields optimal results. In deep learning, model performance is significantly influenced by bias and variance. A model may be too simplistic (high bias) to capture complex patterns in the data, or it may be overly complex (high variance), resulting in overfitting to the training data. Unfreezing 31.25 % of the parameters achieves a balance that is neither too simple nor too complex, thereby enhancing the model's ability to capture data features while preserving its generalization capability. By unfreezing a specific proportion of parameters, the model's capacity for feature representation in the fully connected layers can be increased. A 31.25 % unfreezing ratio may be adequate for the model to learn rich feature representations without causing information redundancy or introducing excessive noise due to an abundance of parameters. The experimental data are presented in Table 4.

4. Results and discussions

4.1. Dataset

This study selected two key datasets, namely the MIMIC-CXR-JPG (Johnson et al., 2019) dataset and the Medical Diff-VQA (Hu et al., 2023) dataset. These two datasets are of significant importances in visual tasks research in the medical domain.

The MIMIC-CXR-JPG dataset is a publicly available collection of chest X-ray images stored in JPG format, accompanied by structured labels derived from the analysis of free-text radiology reports. The primary objective of this dataset is to offer a user-friendly version of MIMIC-CXR while serving as a standard reference for data analysis and image labeling. The dataset comprises 377,110 JPG images, along with structured labels extracted from 227,827 associated radiology reports.

Table 4

Comparison results of unfreezing different numbers of layers in the medical domain visual language generation task.

Methods	B1(%)	R1(%)	RL(%)
MedBLIP-18.75	56.84	61.51	61.10
MedBLIP-25	51.63	56.68	56.37
MedBLIP-31.25	58.95	63.36	63.03
MedBLIP-37.5	53.83	58.87	58.54
MedBLIP-43.75	47.79	54.74	54.64

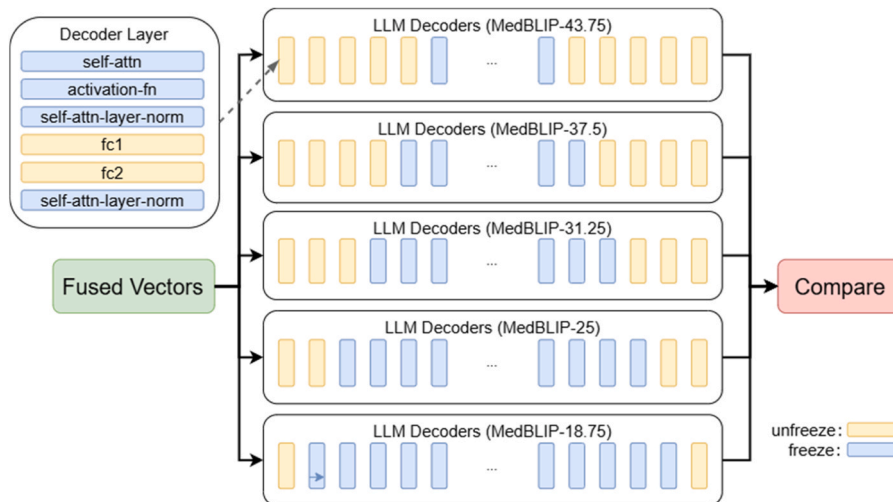


Fig. 4. The unfreezing strategies, depicted from bottom to top in the figure. Illustrate the gradual unfreezing of fully connected layers in the decoder. These strategies correspond to unfreezing proportions of 18.75 %, 25 %, 31.25 %, 37.5 %, and 43.75 %, respectively.

The Medical Diff-VQA dataset is the first dataset specifically designed for the visual question-answering task in the medical domain. These questions are tailored for medical professionals to evaluate diagnoses, interventions, and treatment procedures. The Medical Diff-VQA dataset is derived from the MIMIC-CXR dataset and contains questions categorized into seven distinct groups. In total, the dataset comprises 700,703 question-answer pairs, which are sourced from 164,324 pairs of primary and reference images.

In this study, 53,638 question-answer pairs were initially selected as training data, which was subsequently increased to 70,507 pairs using the previously mentioned mirroring sample generation algorithm. Of these, the training set comprises 49,355 samples, accounting for 70 % of the total dataset. The validation set contains 8045 samples, representing 15 %, while the test set also includes 8045 samples, making up the remaining 15 % of the total dataset, as shown in Table 1.

The selection of these two datasets is crucial for this study, as they offer extensive medical image data along with corresponding question-answer pairs, thereby establishing a robust foundation for model training and evaluation. Furthermore, the structured labels and expertly crafted questions within these datasets allow for a more precise assessment of the model's performance in medical image analysis and question-answering tasks.

4.2. Implementation details

This study employed the Hugging Face Transformers library to load the pre-trained weights of the BLIP-2 model, a robust multimodal model. The research fine-tuned this model for the task of medical visual question answering and utilized an augmented dataset to ensure that the model thoroughly comprehends the visual and linguistic features within the medical domain. This experiment validated the feasibility of enhancing the performance of large-scale models in medical visual question answering tasks through the use of augmented data.

During fine-tuning, the maximum token length was set to 100 to limit the length of input sequences, facilitating easier processing and training of the model. The learning rate was established at $1e-5$, a commonly used value for fine-tuning models. The maximum number of epochs was set to 100 to ensure that the model had sufficient time to converge. The batch size was configured to 4 to balance memory usage and training speed throughout the training process. All model training was conducted on an Nvidia 3090 24 GB GPU, a high-performance graphics processing unit capable of accelerating the model training process.

In this experiment, the model that performed best on the validation set was selected for testing to ensure optimal performance on the validation data. The experiment assessed the model's performance using the BLEU (Papineni et al., 2002) and ROUGE-L (Lin, 2004) metrics. BLEU is a widely used algorithm for the automatic evaluation of machine translation. It measures the correspondence between machine-generated outputs and human-generated outputs, making it a valuable tool for testing the accuracy of generated reports. Its formula is as follows:

$$BLEU = BP \times e^{\left(\sum_{n=1}^N W_n \times \log P_n\right)} \quad (8)$$

P_n is the probability of matching different n grams, calculated by dividing the number of matches by the total number of n grams in the generated text, W_n is the weight of the n -gram, generally set to uniform weight, BP penalty factor. If the length of the translation is less than the length of the shortest reference translation, then BP is less than 1.

ROUGE-L is a widely used metric in machine translation. Based on the statistics of the longest common subsequence, ROUGE-L assesses the relevance between candidate translations and reference documents. Its formula is as follows:

$$R_{lcs} = \frac{LCS(Ref, Answers)}{m} \quad (9)$$

$$P_{lcs} = \frac{LCS(Ref, Answers)}{n} \quad (10)$$

$$ROUGE - L = \frac{(1 + \beta^2) R_{lcs} P_{lcs}}{R_{lcs} + \beta^2 P_{lcs}} \quad (11)$$

Ref represent the standard answers, $Answers$ represent the answers generated by the model. m represents the length of Ref , n represents the length of $Answers$. $LCS(Ref, Answers)$ represents the longest common subsequence of Ref and $Answers$, R_{lcs} represents the recall rate and P_{lcs} represents the precision rate.

Throughout the entire training process, the random seed (12,345) was fixed. This decision was made to ensure that each model was trained in a consistent environment, thereby minimizing the influence of random factors on the experimental results. This approach enhances the reproducibility and comparability of the experiments, ultimately making the results more reliable.

The effects of data augmentation are presented in Table 3. MedBLIP-31.25 involves unfreezing the fully connected layers in the first five and last five decoder layers. In contrast, MedBLIP-31.25-AUG indicates the unfreezing of the fully connected layers in both the first five and last five decoder layers, with its dataset augmented using enhancement algorithms. The MedBLIP-31.25-AUG model outperforms the MedBLIP-31.25 model across all three metrics: B1, R1, and RL. Specifically, the B1 metric improved from 58.95 % to 62.42 %, the R1 metric increased from 63.36 % to 66.21 %, and the RL metric rose from 63.03 % to 65.90 %. These results indicate that the data augmentation algorithm effectively enhanced the model's performance. Here, B1 represents BLEU-1, which measures the BLEU score based on exact matches of individual words. R1 denotes the degree of overlap between individual words and references in ROUGE-1. RL represents ROUGE-L, which assesses the overlap between generated text and references.

Table 4 presents a comparison of the results from three fine-tuning strategies. MedBLIP-18.75 indicates the involvement of 18.75 % of the fully connected layer parameters in the large language model for updates, while MedBLIP-25 corresponds to 25 %, MedBLIP-31.25–31.25 %, MedBLIP-37.5–37.5 %, and MedBLIP-43.75–43.75 %. Based on the data comparison in Table 4, MedBLIP-31.25 achieves the best performance, demonstrating that unfreezing the fully connected layers in the first five and last five decoder layers yields optimal results.

In the medical field, a comparison of various fine-tuning strategies for visual language generation tasks revealed that the MedBLIP-31.25 model achieved the highest performance by unfreezing 31.25 % of the parameters in the fully connected layers. Its B1, R1, and RL metrics reached 58.95 %, 63.36 %, and 63.03 %, respectively, surpassing all other models in the comparison. However, increasing the number of unfrozen layers does not always result in improved performance; for instance, the MedBLIP-18.75 model outperformed the MedBLIP-25 model, while the MedBLIP-31.25 model further enhanced performance. This indicates that, for this specific task, a 31.25 % ratio of unfrozen layers is optimal. Notably, when the proportion of unfrozen layers was increased to 37.5 % and 43.75 %, the model's performance actually declined, suggesting that excessive unfreezing may lead to overfitting or optimization challenges for the model.

Table 5

Comparison with other results in medical domain visual language generation tasks.

Methods	B1(%)	R1(%)	RL(%)
BLIP_VQA_BASE	22.28	23.04	23.02
BLIP_VQA_CAPFILT	22.29	23.05	23.03
VILT_b32	26.00	26.43	26.43
GIT_VQA	26.14	27.03	27.03
MedBLIP-31.25-VIT (Ours)	58.95	63.36	63.03
MedBLIP-31.25-AUG-VIT(Ours)	62.42	66.21	65.90
MedBLIP-31.25-AUG-VC(Ours)	62.10	67.30	66.12

Table 5 presents the performance of MedBLIP on the medical visual language dataset. The MedBLIP-31.25-VIT model utilizes a transformer-based image extractor and unfreezes 31.25 % of the fully connected layers in the large language model. Building on this foundation, the MedBLIP-31.25-AUG-VIT model further enhances performance by incorporating a mirror sample augmentation algorithm. Additionally, the MedBLIP-31.25-AUG-VC model introduces a multi-layer convolutional image feature extractor on top of the MedBLIP-31.25-AUG-VIT, thereby improving image processing capabilities. According to the comparative results in Table 5, the proposed MedBLIP-31.25-VIT, MedBLIP-31.25-AUG-VIT and MedBLIP-31.25-AUG-VC models excel in medical visual language generation tasks, outperforming other visual language models, including BLIP_VQA_BASE, BLIP_VQA_CAPFILT (Li et al., 2022), VILT_b32 (Kim et al., 2021) and GIT_VQA (Wang et al., 2022) models. Among all the comparison models, the MedBLIP-31.25-AUG-VC model demonstrated the highest performance, achieving B1, R1, and RL metrics of 62.10 %, 67.30 %, and 66.12 %, respectively. In this experiment, we calculated the confidence intervals for the final model, MedBLIP-31.25-AUG-VC. The results show that the confidence interval for the B1 metric is [62.10, 63.85], for the R1 metric it is [67.01, 68.68], and for the RL metric, it is [66.67, 68.38]. The confidence interval demonstrates the stability of the experimental results. Notably, the performance advantage of the MedBLIP model is attributed to the incorporation of a multi-layer convolutional image feature extractor. This finding suggests that the experimental approach offers significant benefits in addressing medical visual language generation tasks, opening new possibilities for related applications in healthcare.

As shown in Table 6, this study utilizes the results for statistical analysis by calculating p value between MedBLIP-31.25-VIT and the MedBLIP-31.25-AUG-VIT, as well as between the MedBLIP-31.25-AUG-VIT and the MedBLIP-31.25-AUG-VC. The results of p value are less than 0.05, indicating that the mirror sample generation technology and the image feature extractor significantly enhance the model's performance.

The cosine similarity (Kallus, 2021) between the weights after fine-tuning and the original weights serves as a method for measuring the differences between these two sets of weights. Cosine similarity assesses the similarity between two vectors by calculating the cosine of the angle between them. If the directions of the two vectors are identical, the cosine similarity is 1; if they are completely opposite, it is -1 ; and if they are orthogonal, it is 0. In Fig. 5, the cosine similarity between the weights of the first four layers and the last four layers of the large language model after fine-tuning, compared to the corresponding weights of the native model, is calculated and represented in a two-dimensional coordinate system. The green area indicates that the weights of the model before and after fine-tuning are essentially the same, while the red area signifies that the weights have undergone significant changes. This visualization facilitates an intuitive understanding of the extent of weight alterations during fine-tuning. If the cosine similarity of a pair of weights exceeds a specified threshold, it indicates substantial differences between the weights after fine-tuning and the original weights. Such changes suggest that the model has acquired new knowledge during fine-tuning, particularly in relation to the medical data. In other words, the model 'remembers' certain features from the medical data, which enhances its performance in the medical domain.

Fig. 6 illustrates the final question-and-answer results of the MedBLIP model. Panels (a)–(f) display chest X-ray images corresponding to different question types, where Q represents the patient's question related to the image, and A denotes the answer generated by the model.

Table 6
Statistical significance tests by p value.

Methods	p value (B1)	p value (R1)	p value (RL)
MedBLIP-31.25-AUG-VIT	1.05×10^{-4}	5.80×10^{-4}	6.52×10^{-4}
MedBLIP-31.25-AUG-VC	2.38×10^{-18}	7.64×10^{-17}	7.18×10^{-17}

By inputting the chest X-ray images along with their corresponding questions into the MedBLIP-31.25-AUG-VC model, the model provides diagnoses based on the images and questions, as depicted in Fig. 4. The experimental results demonstrate the effectiveness of MedBLIP-31.25-AUG-VC in distinguishing between anterior and anteroposterior positions in panel (a), identifying location issues in panels (b) and (e), addressing abnormality concerns in panels (c) and (d), and resolving level issues in panel (f).

As shown in Table 7, the model has demonstrated the ability to detect signs of abnormalities and identify key issues in the images. However, compared to human radiologists, the model's output is more concise and lacks detailed analysis. Human radiologists offer more comprehensive and specific diagnostic information in their reports. Although the model cannot fully replicate the complexity and detail of human radiology experts, it still plays a significant role in the diagnostic process. Firstly, the model can rapidly identify abnormal signs in medical images, providing radiologists with preliminary screening results that help accelerate diagnosis. Secondly, the model aids in decision-making; its output serves as a reference for radiologists, offering them a preliminary understanding of potential diagnostic directions before they conduct a detailed analysis. In situations where medical resources are limited, the model assists radiologists in managing large volumes of imaging data, thereby optimizing their workload and allowing them to devote more time and effort to complex cases that require professional judgment. Additionally, the model provides standardized outputs, which help reduce diagnostic inconsistencies arising from variations in individual radiologists' experiences. Finally, the model can be utilized as an educational tool, helping medical students and radiologists learn to identify specific imaging features, thereby enhancing their diagnostic skills.

Table 8 illustrates that "Q" denotes user questions, "A" signifies model outputs, and "R" indicates reference answers. When inquired about the severity of a medical condition, the model's responses were unsatisfactory. This may be attributed to the model's inability to account for the dynamic changes associated with the condition. The severity of a medical condition can fluctuate over time; however, the model may have solely relied on static image data, neglecting the temporal changes in the time series. Consequently, it failed to recognize the gradual worsening trend of the condition.

This study utilizes the VQA-RAD dataset (Lau et al., 2018) including both CT and MRI to assess generalization of MedBLIP based on different imaging modalities. As shown in Table 9, the results after training indicate that 31.25 % of the unfrozen proportion of the MedBLIP architecture demonstrates a certain degree of generalization ability for MRI, CT, and other images. Additionally, the evaluation results of our model are slightly higher than the BLIP-2 model aiming at the VQA-RAD dataset. This also demonstrates that the model proposed exhibits strong generalization capabilities.

4.3. Future work and limitations

Our current research focuses on the application of chest X-ray data, specifically utilizing the MIMIC-CXR-JPG and Medical Diff-VQA datasets. These datasets were selected due to their widespread use in this field and the high-quality annotations associated with relevant tasks. Although our dataset also includes images from other imaging modalities (e.g., MRI, CT), the absence of high-quality question-answer annotations for these modalities limits the generalizability of our findings in this study. In future work, we plan to integrate more resources to extend the model to encompass other imaging modalities and non-thoracic domains, while exploring cross-modal learning approaches to enhance generalization capabilities.

We acknowledge that using BLEU and ROUGE as evaluation metrics for translation quality has certain limitations. These metrics may not fully capture the semantic content, style, and contextual appropriateness of translations. Therefore, while our findings provide a useful reference,

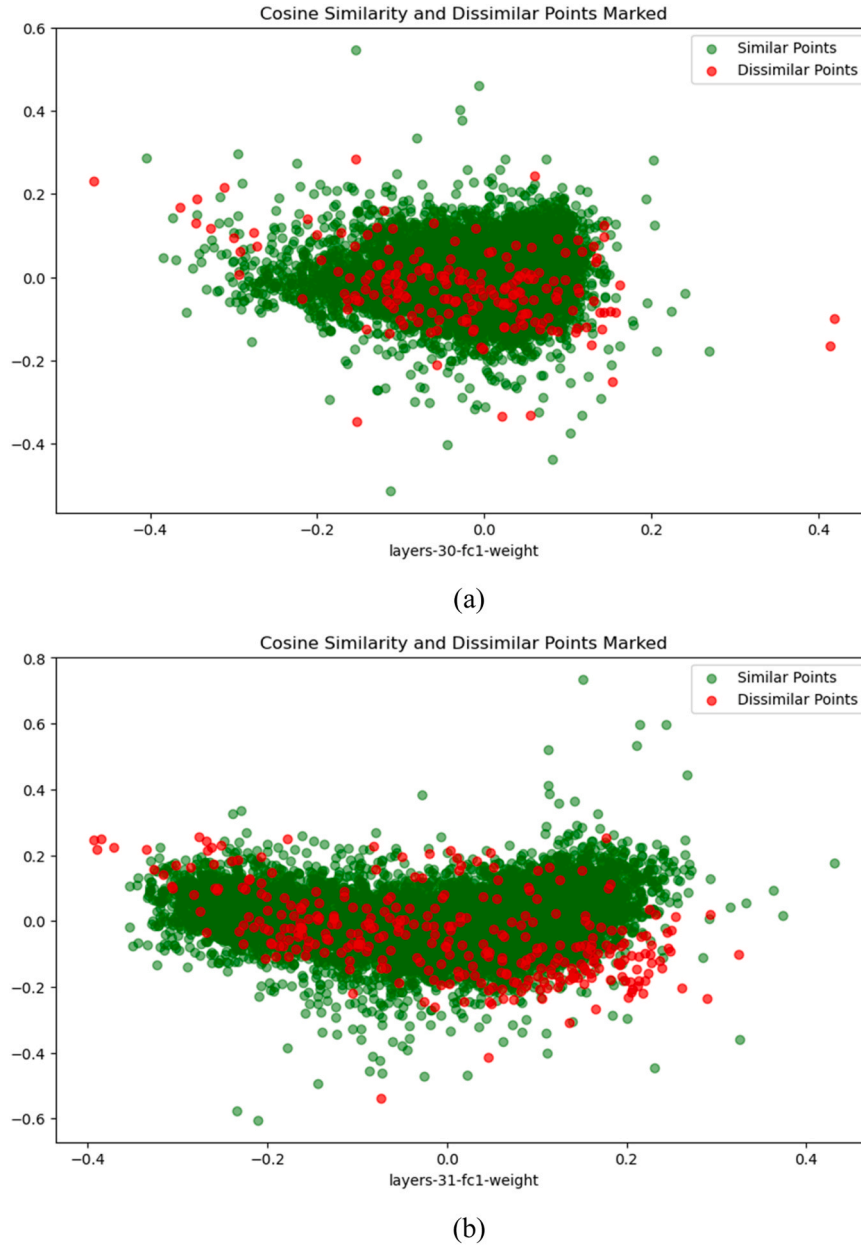


Fig. 5. Comparison of cosine similarity for layers 30 (a) and 31 (b) in the large language model. In the chart, the x-axis and y-axis represent the values obtained by mapping the weights to two dimensions, with red indicating significant changes in values after fine-tuning.

they should not be regarded as the sole measures of translation quality. Future research could incorporate additional evaluation methods, such as human assessments and deep learning-based approaches, to achieve a more comprehensive and accurate evaluation of translation quality.

Future directions involve further optimizing model performance, expanding to additional medical application scenarios, and continuously enhancing the accuracy and efficiency of the model in medical image and text question-answering tasks. This innovative approach opens new avenues for development in the healthcare sector and offers robust support for the realization of intelligent medical question-answering systems.

5. Conclusions

In this study, we propose a multi-modal medical visual-language question-answering model named MedBLIP. This method employs a mirrored sample generation algorithm to enhance the medical question-

answering dataset, thereby improving data diversity and augmenting the model's generalization capabilities. Additionally, we introduce a multi-layer convolutional image feature extractor to enhance the model's comprehension of medical images and its adaptability to various tasks. Finally, by fine-tuning the fully connected layers of the language model, this work unfreezes specific portions of the large language model to generate more accurate diagnoses. Among all the models compared, the MedBLIP-31.25-AUG-VC model demonstrated the highest performance, achieving BLEU-1, ROUGE-1, and ROUGE-L metrics of 62.10 %, 67.30 %, and 66.12 %, respectively.

In the experimental section, the MedBLIP model is compared with other visual-language models, demonstrating its superior performance in medical visual-language generation tasks. The effectiveness of the multi-layer convolutional image feature extractor is validated through experiments, and five unfreezing strategies are employed to identify effective fine-tuning methods for large language models. The experimental results confirm the feasibility of visual-language models in

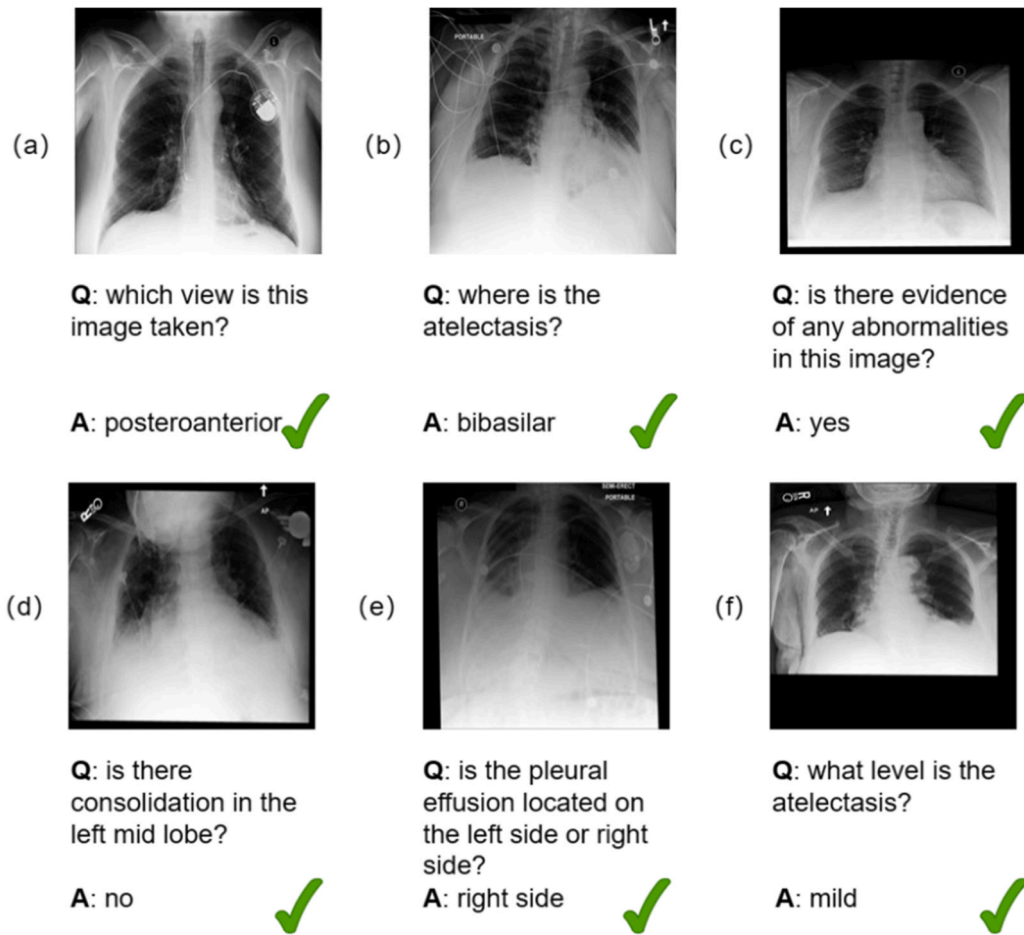


Fig. 6. Results of MedBLIP-31.25-AUG-VC for different question types in the medical visual language generation task.

Table 7

Comparison of model-generated reports and human expert reports.

MedBLIP	Human Radiologists
Q: is there evidence of any abnormalities in this image? A: yes Q: where in the image is the lung opacity located? A: bilateral	The endotracheal tube and nasogastric tubes have been removed. Following chest tube removal, there is no definite evidence of pneumothorax. <u>Diffuse bilateral pulmonary opacifications persist, though they may be somewhat decreased since the previous study.</u> The appearance probably reflects a combination of pulmonary edema and multifocal pneumonia. There is rounded opacity in the left mid lung likely <u>atelectasis</u> , and there is veil like opacity over the left lung which is likely layering left pleural effusion. The <u>small</u> right pleural effusion has increased and increasing opacity in the right lower lobe, likely worsening atelectasis.
Q: is there atelectasis? A: yes Q: what level is the pneumothorax? A: small	

addressing medical question-answering tasks.

Our current research focuses on the application of chest X-ray data, with plans to expand to other imaging modalities and non-thoracic areas in the future. Additionally, we aim to explore cross-modal learning to enhance the model's generalization capabilities. We acknowledge the limitations of existing evaluation metrics and will introduce more comprehensive evaluation methods in future studies. This research will continue to optimize model performance, broaden its medical application scenarios, and improve the accuracy and efficiency of medical image question-answering tasks, thereby supporting the development of intelligent medical question-answering systems.

MedBLIP, a medical visual question-answering method based on

Table 8

Model error analysis.

(Cases) MedBLIP	Reference answers	Error Analysis
Q:what level is the edema? A:mild	R:increasing mild	The model may not have accurately captured the dynamic changes associated with edema. The degree of edema can fluctuate over time, and the model may have relied solely on static image data, neglecting the temporal changes in the time series. This oversight could lead to a failure in identifying the increasing trend of edema.
Q:what level is the atelectasis? A:mild	R:severe	The model may have significantly underestimated the severity of atelectasis. The severity of atelectasis is usually determined by imaging examinations, and the model may have had significant errors in processing these imaging data, leading to inaccurate judgment of the severity of atelectasis.

Table 9

Generalization of MedBLIP.

Methods	B1 (%)	R1 (%)	RL (%)
BLIP-2 (Li et al., 2023)	43.97	45.49	45.48
MedBLIP-31.25-AUG-VC	46.35	46.80	46.80

large language models, theoretically offers promising diagnostic outcomes, assisting physicians in making faster and more accurate decisions. However, the implementation of MedBLIP in real-world hospital systems still encounters several challenges. In terms of computational resources, MedBLIP necessitates robust computing capabilities, including high-performance GPUs and substantial memory, which may restrict its implementation in certain hospitals. With respect to data security, medical data is highly sensitive; therefore, the deployment of MedBLIP must prioritize data protection to prevent leaks and misuse. To keep the model up to date, a significant volume of medical domain data is required for incremental training. However, acquiring this data can be challenging and involves extensive preprocessing tasks, such as data cleaning and annotation. To address these challenges, the following solutions could be considered. Deploying MedBLIP on cloud platforms to leverage scalable computing resources and streamline model updates and maintenance, and establishing a medical data-sharing platform to simplify data access for researchers seeking to train models.

The model can assist radiologists by playing a significant role in preliminary screening and highlighting potential issues, which helps radiologists focus on key areas more efficiently. However, the model is not yet capable of replacing radiologists. The professional knowledge and experience of radiologists are essential for interpreting images, making accurate diagnoses, and providing treatment recommendations. Therefore, the model should be viewed as an auxiliary tool designed to enhance the efficiency of radiologists rather than replace them.

CRedit authorship contribution statement

Yimu Ji: Writing – review & editing, Validation. **Lejun Gong:** Writing – original draft, Conceptualization. **Jiaming Yang:** Writing – original draft, Software. **Shengyuan Han:** Formal analysis, Data curation.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in the manuscript entitled “**MedBLIP: A Multimodal Method of Medical Question-Answering Based on Fine-Tuning Large Language Model**”.

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Data availability

Data will be made available on request.

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